# Exit Ticket · reflect and retrieve

## Exit Ticket

A linked list always offers O(1) indexed access to its nth node.

## Use key vocabulary from memory

1. Explain node in your own words.

2. Explain Next in your own words.

3. Explain Previous in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.





# Do Now · retrieval

## Lesson 051 · Linked Lists: Nodes and References

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

How does Queue<T> remove items?

### Q2

How does a stack differ from a queue?

### Q3

What is an object reference?

### Q4

A singly linked list is HEAD → node5 → node9 → NULL. A new node7 is available. Describe the two next-pointer assignments needed to insert node7 immediately after node5 without losing access to node9.

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

### Core idea 1

Discover — A linked-list node holds a value plus one or more links to neighbouring nodes, rather than occupying an indexed fixed-size array slot.

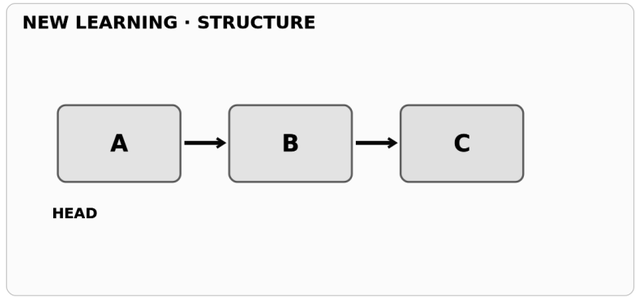
### Core idea 2

Trace — C# LinkedList<T> is doubly linked: nodes expose Previous and Next; AddAfter inserts relative to an existing node.

### Core idea 3

Evaluate — To insert B between A and C, change logical links to A↔B↔C. The operation differs from shifting array elements.

**Key visual · complete concept**




### Annotate the key visual

Explain which labels or features show the main process.





# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

### Question 3

Compare middle insertion in linked list with a C# List<int>, including locating position and access. [6]

Check that you have used the technical terms and explained your decisions.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Draw the logical next-links in HEAD→3→6→NULL after inserting 4 after 3. [6]

## Question 2

Write C# using LinkedList<int> to create 2,4 then insert 3 between them. [6]

Check that you have used the technical terms and explained your decisions.





# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

node — Value and link information.

Next — Reference to following node.

Previous — Reference to preceding node.

head — First list node.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

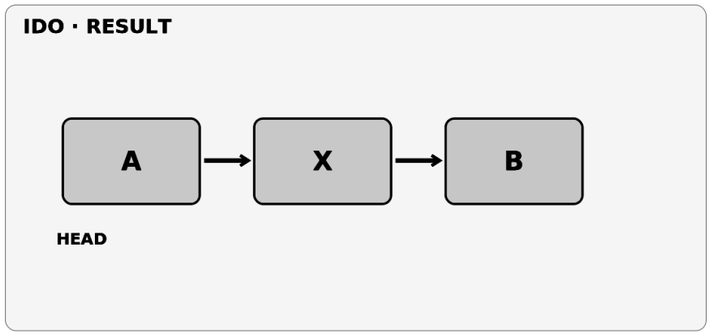
## Problem

Trace AddAfter to insert X after A in A→B.

## Follow the worked method

1. Explain the trace — New order A,X,B. The inserted node links between A and B.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Explain the trace
P,R remain. After deletion P and R are adjacent logical neighbours.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.